Bounds and Estimators for the Civilian-to-Combatant Death Ratio from Aggregation of Conflicting Sources

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Abstract

Casualty figures in armed conflict are reported by sources with opposing incentives: belligerents inflate enemy combatants killed and deflate civilians; advocacy and intergovernmental bodies push the other way. We treat the civilian-to-combatant death ratio as a partially identified parameter and ask what its value can be when conflicting sources are aggregated. The paper delivers two objects.

(1) Bounds from under-fudgible relations. Under a one-parameter model in which civilian adult men face up to $\bar{\mu}$ times the death rate of women and children, we give a sharp identified set for the combatant share $q = D_M/D$ as a function of population structure, the demographic composition of the dead, and pre-war combatant stock; intersected with the mechanical bound $D_M \leq M$, this is a closed-form interval. Because the relation is under-fudgible—moving any one input forces a compensating move in independently measured ones—we define the *contradiction radius* ρ: the smallest standardised perturbation of an independent input that makes a disputed claim feasible.

(2) Confidence coefficients under political bias. Each source is allowed to be Huber ε_i-contaminated; the analyst's political-bias prior on a source enters as ε_i. We give a closed-form bias-robust credible interval that absorbs the assumed contamination, with a sharper quantile-displacement bound under a density lower bound at the quantiles.

In the Gaza 2023–2026 application, aggregating Palestinian census, OHCHR identifications, ministry-of-health records, and independent manpower estimates, the public Israel Defense Forces (IDF) figure of 17–25k combatants killed has $\rho \gtrsim 20$. A spatial Bayesian model on the same inputs yields $q = 2.5\%$ with a 95% credible interval of $[1.6\%, 3.8\%]$ and an implied civilian-to-combatant ratio of 43:1 $[30, 64]$. The naive "uniform random violence" null is rejected at $p \approx 1.7 \times 10^{-16}$, but rejecting it is necessary, not sufficient, to rationalise any large combatant share.

1 Introduction

1.1 The aggregation problem

Casualty figures in armed conflict are reported by sources with opposing incentives. Belligerents inflate enemy combatants killed and deflate civilians; advocacy organisations, opposing belligerents, and intergovernmental bodies have symmetric incentives in the other direction. Naive aggregation is therefore biased in a direction that depends on which side the analyst trusts more. The tempting conclusion is that civilian-to-combatant ratios are too politicised to analyse statistically.

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We argue for a narrower position. Conflicting source aggregates fail in well-understood ways, but the parameters they purport to measure are jointly constrained by a small number of relations that are difficult to manipulate. An analyst who treats every belligerent source as adversarial can still recover an informative bound by exploiting these constraints; what cannot be recovered without further assumptions is the position of the truth *within* the bound. This separation—bounds from data, position from politics—is the organising principle of the paper.

The closest precedents are Spagat et al. (2009) on the contestation of war-death estimates, Lum et al. (2013) on multiple systems estimation under reporting failure, Radford et al. (2023) on Bayesian inference of conflict losses with reporting biases, Vesco et al. (2026) on probabilistic UCDP underreporting, and Gómez-Ugarte et al. (2025) on uncertainty quantification for Gaza mortality. Methodologically we draw on partial identification (Manski, 2003, 2007; Molinari, 2020; Kline and Tamer, 2023) and robust Bayesian analysis (Huber, 1964, 1981; Berger and Berliner, 1986).

1.2 Two deliverables

Bounds. Where can the civilian-to-combatant ratio possibly lie, given a set of independently measured demographic and manpower facts? The answer is a sharp identified set in the sense of Manski's partial-identification programme (Manski, 2003; Tamer, 2010; Romano and Shaikh, 2010). If the identified set is narrow, no source-aggregation rule can pull the truth outside it. The width is determined by the data, not by the analyst's politics.

Confidence coefficients. How much can the analyst's political-bias prior on each source widen this set? Each source is Huber ε-contaminated (Huber, 1981; Berger and Berliner, 1986): it is true with probability $1 - \varepsilon$ and adversarial with probability ε. A bias-robust credible interval inflates by $\sum_i \varepsilon_i R_i$, where R_i is the diameter of the identified set when source i is removed. Setting $\varepsilon_i \approx 1$ for belligerent claims and $\varepsilon_i \approx 0.05$ for census-grade sources gives an interval honest about which inputs are doing the work.

The two objects are complementary: the first is information that survives any reasonable politics; the second is the rate at which uncertainty grows as the analyst dials in distrust of specific sources.

1.3 The under-fudgible mechanism

The bounds are not vacuous because of demographic arithmetic. If a source claims many combatants were killed, then either (i) the identified dead contain fewer women and children, (ii) the population contains fewer women and children, (iii) the armed group had a larger pre-war stock, or (iv) civilian adult men were killed at a much higher per-capita rate than civilian women and children. These quantities are measured by *different* institutions: a national or UN census produces (ii); the Office of the High Commissioner for Human Rights (OHCHR), the International Committee of the Red Cross (ICRC), or a local health ministry produces (i); independent intelligence services produce (iii); (iv) is bounded by exposure-time studies. They are not equally manipulable.

Definition 1.1 (Under-fudgible relation)**.** A relation $R(x_1, \ldots, x_K) = 0$ between conflict statistics is *under-fudgible* if each x_k is reported by several non-aligned institutions or has uncontroversial physical or demographic bounds, and violating R in any single coordinate forces a deviation of several measured standard errors in at least one of the others.

The simplest such relation is the manpower bound $D_M \leq M$. The more informative one is demographic: if w is the women-and-children share of the population and ω their share among the dead, then high combatant-share claims force ω down. When ω is observed to be high, a high-D_M claim must be paid for either by a very large adult-male exposure multiplier or by contradicting the population baseline. We measure this trade-off as a *contradiction radius* ρ: the smallest standardised perturbation of any independent input that brings a disputed claim back into the feasible region.

1.4 Contributions and roadmap

(i) A sharp identified set for $q = D_M/D$ under bounded differential exposure of civilian adult men (Theorem 3.4), intersected with the manpower bound (Corollary 3.5).

(ii) A precision-weighted aggregation rule for multiple demographic sources, with delta-method standard error and contamination-aware bias bound (Proposition 2.1).

(iii) A feasibility test for any aggregated claim, stated as a distance-to-feasibility optimisation: the contradiction radius ρ (Definition 4.2, Theorem 4.3).

(iv) A bias-robust credible interval under Huber ε-contamination, with a sharper quantile-displacement bound (Proposition 5.1).

(v) A worked Gaza 2023–2026 application aggregating six independent source channels, plus a supplement applying coarser diagnostics to 81 conflicts from 1900–2026.

Sections 2–4 develop the bounds; Section 5 adds the confidence-coefficient layer; Section 6 runs both on Gaza; Section 7 surveys 81 conflicts. The method does not assign moral responsibility, estimate indirect deaths, or replace forensic work.

2 Setup

A conflict produces $D \in \mathbb{N}$ deaths in $[t_0, t_1]$ in a population of size N. Each death is a combatant ($C = 1$) or civilian ($C = 0$); write $D_M = \sum_i \mathbf{1}\{C_i{=}1\}$, $D_C = D - D_M$, and the targets

$$q = D_M/D \in [0, 1], \qquad r = D_C/D_M = (1 - q)/q.$$

Each death has a demographic class $k \in \mathcal{K} = \{\text{child}, \text{adult-female}, \text{adult-male}\}$. Write $\omega_k = D_k/D$ for the share among the dead, p_k for the share in the population, and let AM denote adult male (ages 18–60), with population share $a = p_{\mathrm{AM}}$. Let $w = \sum_{k \neq \mathrm{AM}} p_k$ be the women-and-children population share and $\omega = \sum_{k \neq \mathrm{AM}} \omega_k$ their share of the dead. Let M be the pre-war combatant stock and $f = M/N$. Without loss combatants are in AM; female and minor combatants enter as a small slack $\bar{\eta} \leq 0.03$ (Remark 3).

Table 1. Core notation and measurement channels.

Symbol	Meaning	Role	Typical source
D	total deaths in window	input	ministry of health, OCHA, UCDP, survey
D_M, D_C	combatant, civilian deaths	target	decomposition of D
$q = D_M/D$	combatant share of dead	estimand	—
w	women+children pop. share	input	census, UN Population Division
$a = p_{\mathrm{AM}}$	adult-male pop. share	input	census / age pyramid
ω	women+children share of dead	input	identified-deaths sample (e.g. OHCHR)
M	combatant stock at t_0	input	IISS / RUSI / CSIS / intelligence
$\bar{\mu}$	exposure-differential cap	assumption	sensitivity parameter
ρ	contradiction radius	output	distance to feasibility

Sources. We use five channels: an independent population census (p_k, N); a demographic survey of the dead returning sample shares $\hat{\omega}_k$; aggregate death totals $\hat{D}$; belligerent combatant-death claims $\hat{D}_M$; and independent manpower estimates $\hat{M}$. Each source i is allowed to be ε_i-contaminated in Huber's sense (Huber, 1964; Berger and Berliner, 1986): it draws from the truth with probability $1 - \varepsilon_i$ and from an arbitrary adversary with probability ε_i.

2.1 Source aggregation

When several independent sources $i = 1, \dots, m$ report the same scalar parameter θ (e.g., the women-and-children share ω from OHCHR identifications and from MoH death records) with reported point estimates $\hat{\theta}_i$ and standard errors $\hat{\sigma}_i$, the natural *precision-weighted* aggregate is

$$\hat{\theta}_{\text{agg}} = \frac{\sum_i \hat{\theta}_i/\hat{\sigma}_i^2}{\sum_i 1/\hat{\sigma}_i^2}, \qquad \hat{\sigma}_{\text{agg}}^2 = \Big(\sum_i 1/\hat{\sigma}_i^2\Big)^{-1}. \tag{1}$$

Proposition 2.1 (Aggregation under conflicting sources)**.** *If sources are independent and unbiased, $\hat{\theta}_{\text{agg}}$ in* (1) *is the minimum-variance unbiased linear estimator of θ. If source i has bias b_i, the aggregate has bias $(\sum_i b_i/\hat{\sigma}_i^2)/(\sum_i 1/\hat{\sigma}_i^2)$. Allowing ε_i-contamination (Section 5), the worst-case bias of $\hat{\theta}_{\text{agg}}$ is at most $\sum_i \varepsilon_i R_i \cdot \hat{\sigma}_{\text{agg}}^2/\hat{\sigma}_i^2$, where R_i is the diameter of the support of source i's contaminating distribution.*

Proof. Inverse-variance weighting minimises variance over linear unbiased combinations (Gauss–Markov). The bias and contamination statements follow by linearity. □

In the Gaza application, the OHCHR identification sample ($\hat{\omega} = 0.693$) and the MoH death records ($\hat{\omega} = 0.560$) are aggregated; both have estimated $\hat{\sigma} \approx 0.01$, giving the geometric blend $\hat{\omega}_{\text{agg}} \approx 0.620$ used throughout. The corresponding contradiction radius is reported separately for each anchor and for the blend, so the reader can see how aggregation choices propagate.

3 Bounds: partial identification of q

This section delivers a sharp identified set for q that any source-aggregation rule must respect. We begin with a benchmark assumption, then weaken it.

Assumption 3.1 (Exchangeable collateral)**.** Civilians of all demographic classes face equal per-capita death hazard.

Theorem 3.2 (Demographic identification)**.** *Under Assumption 3.1,*

$$q = 1 - \frac{\omega(1-f)}{w}. \tag{2}$$

Proof. Civilian deaths split between $W = \{k \neq \text{AM}\}$ and AM in proportion to their civilian-population shares, $w/(1-f)$ and $(a-f)/(1-f)$. Combatant deaths fall entirely in AM. Hence $\omega = (1-q)w/(1-f)$, which inverts to (2). □

Remark. A small slack $\bar{\eta} \leq 0.03$ for combatants outside AM shifts (2) by $-\bar{\eta}$ and propagates mechanically through what follows; we suppress it.

Assumption 3.1 is strong: civilian adult men typically face higher per-capita exposure than women and children due to outside work, food queues, rescue activity, and circumstantial proximity to combat. We weaken it to a one-parameter family.

Assumption 3.3 (μ-bounded differential exposure)**.** There exists $\mu \in [1, \bar{\mu}]$ such that, conditional on being civilian, class AM has per-capita death rate μ times that of class W.

Theorem 3.4 (Sharp identified set)**.** *Suppose $f \le a$ and Assumption 3.3 holds. Given $(\omega, w, a, f, \bar{\mu})$, the identified set for q is the image of the monotone-decreasing curve*

$$q(\mu) = 1 - \omega\,\frac{w + \mu(a - f)}{w}, \qquad \mu \in [1, \bar{\mu}], \tag{3}$$

giving the sharp interval $[q(\bar{\mu}),\, q(1)] \cap [0, 1]$.

Proof. Let λ be the civilian per-capita death hazard in W. Expected civilian deaths are proportional to $\lambda w N + \lambda\mu(a - f)N$, so the civilian share landing in W is $w/(w + \mu(a - f))$. Since only the civilian fraction $1 - q$ of deaths reaches W, $\omega = (1 - q)w/(w + \mu(a - f))$. Solving gives (3); $\partial_\mu q = -\omega(a - f)/w \le 0$. Sharpness: every $\mu \in [1, \bar{\mu}]$ corresponds to a feasible hazard pair $(\lambda, \mu\lambda)$ that attains $q(\mu)$. □

Corollary 3.5 (Manpower bound)**.** *$q \le M/D$, so the joint feasible interval is*

$$q \in \big[\max\{0,\, q(\bar{\mu})\},\; \min\{1,\, q(1),\, M/D\}\big].$$

Plug-in estimator and standard error. The sample analogue of the upper endpoint is $\hat{q} = 1 - \hat{\omega}(1 - \hat{f})/\hat{w}$, with delta-method variance

$$\widehat{\mathrm{Var}}(\hat{q}) \approx \frac{\hat{\omega}^2}{\hat{w}^2}\widehat{\mathrm{Var}}(\hat{f}) + \frac{(1 - \hat{f})^2}{\hat{w}^2}\widehat{\mathrm{Var}}(\hat{\omega}) + \frac{\hat{\omega}^2(1 - \hat{f})^2}{\hat{w}^4}\widehat{\mathrm{Var}}(\hat{w}).$$

Gaza-implied range. With $w = 0.733$, $\hat{\omega} = 0.620$ (geometric mean of OHCHR 0.693 and MoH-record 0.560), $a = 0.255$, $\hat{f} = 0.020$, $\hat{M} = 45{,}000$, $\hat{D} = 70{,}000$, and $\bar{\mu} = 2.5$, Corollary 3.5 gives $q \in [0\%, 17\%]$; the manpower bound is non-binding. The spatial Bayesian model of Section 6 concentrates inside this interval at $q \approx 2.5\%$.

4 Bounds: contradiction radius

Theorem 3.4 returns an interval; a public claim returns a point. The diagnostic measures their distance after standardising each input by its measurement uncertainty.

Definition 4.1 (Feasible set)**.** For inputs $x = (\omega, w, a, f, M, D)$ and exposure cap $\bar{\mu}$,

$$\mathcal{F}(x; \bar{\mu}) = \Big\{ q \in [0, 1] :\ \exists\, \mu \in [1, \bar{\mu}] \text{ with } q = 1 - \omega\tfrac{w + \mu(a - f)}{w} \text{ and } qD \le M \Big\}. \tag{4}$$

A claim $\hat{D}_M^{(i)}$ converts to $q^{(i)} = \hat{D}_M^{(i)}/\hat{D}$ and is feasible iff $q^{(i)} \in \mathcal{F}(\hat{x}; \bar{\mu})$.

Definition 4.2 (Contradiction radius)**.** Let $\hat{x} = (\hat{\omega}, \hat{w}, \hat{a}, \hat{f}, \hat{M}, \hat{D})$ with measured standard errors $\sigma_\omega, \sigma_w, \sigma_M$. The contradiction radius of claim $q^{(i)}$ is

$$\rho^{(i)} = \inf_{\omega', w', M'} \max\Big\{ \tfrac{|\omega' - \hat{\omega}|}{\sigma_\omega},\ \tfrac{|w' - \hat{w}|}{\sigma_w},\ \tfrac{(M' - \hat{M})_+}{\sigma_M} \Big\} \quad \text{s.t.} \quad q^{(i)} \in \mathcal{F}((\omega', w', \hat{a}, \hat{f}, M', \hat{D}); \bar{\mu}). \tag{5}$$

Theorem 4.3 (Feasibility and contradiction)**.** *With $(\hat{a}, \hat{f}, \hat{D}, \bar{\mu})$ fixed, a claim is consistent with the independent inputs iff $\rho^{(i)} = 0$. If $\rho^{(i)} > 0$, every rationalisation moves at least one of ω, w, M by $\rho^{(i)}$ standard-error units.*

Proof. $\mathcal{F}$ is the intersection of the identified set from Theorem 3.4 with $qD \le M$, and is closed in (ω, w, M). The infimum in (5) is zero iff the observed inputs themselves are feasible, by closedness. Otherwise, the ℓ_∞ objective bounds every coordinate move from below. □

Computation. Because $q(\mu)$ is monotone, the boundary of $\mathcal{F}$ is attained on one of three axes: the demographic axis (ω moves), the population-composition axis (w), or the manpower axis (M/D). This is the contradiction triangle of Figure 2; (5) is a low-dimensional convex programme.

Diagnostic procedure.

D1. Fix the analysis window and total deaths D.

D2. Estimate (w, a) from the population baseline and ω from an identified-deaths sample.

D3. Choose $\bar{\mu}$ and an independent upper bound on M.

D4. Compute $\mathcal{F}(\hat{x}; \bar{\mu})$ from (4).

D5. For every disputed $\hat{D}_M^{(i)}$, report $q^{(i)} = \hat{D}_M^{(i)}/D$, the membership $q^{(i)} \in \mathcal{F}$, and $\rho^{(i)}$.

The test is symmetric: it rejects "too many combatants" and "zero combatants" against the same set.

5 Confidence coefficients under political bias

The bounds in Sections 3–4 treat each measured input as honest. We now allow the analyst to dial in distrust of specific sources. Political bias enters the confidence statement through Huber contamination: each source i writes its likelihood as $(1 - \varepsilon_i)L_i + \varepsilon_i Q_i$, with Q_i adversarial in some class $\mathcal{Q}$ (Huber, 1964, 1981; Berger and Berliner, 1986; Gagnon, 2023). The contamination parameter $\varepsilon_i \in [0, 1]$ is the analyst's prior probability that source i is producing a politically motivated reading rather than a measurement of the true distribution.

Proposition 5.1 (Bias-robust credible interval)**.** *Let θ be a scalar estimand with $1-\alpha$ posterior credible interval $I_0 = [L_0, U_0]$ under face-value sources. Suppose each source is ε_i-contaminated and removing source i yields an identified set for θ of diameter R_i. Then*

$$I_\varepsilon \;=\; \Big[L_0 - \textstyle\sum_i \varepsilon_i R_i,\; U_0 + \sum_i \varepsilon_i R_i\Big] \tag{6}$$

contains every $1 - \alpha$ credible interval obtainable by adversarial replacement of each L_i in $\mathcal{Q}$. If the marginal density of θ exceeds $m > 0$ near each quantile, the sharper quantile displacement

$$\big|\hat{Q}_\varepsilon(u) - \hat{Q}_0(u)\big| \;\le\; m^{-1}\big(1 - \textstyle\prod_i(1 - \varepsilon_i)\big), \qquad u \in \{\alpha/2,\, 1 - \alpha/2\},$$

also holds.

Proof. Expanding $\prod_i\big[(1 - \varepsilon_i)L_i + \varepsilon_i Q_i\big]$ as a mixture, the total mass on components in which source i is contaminated is at most ε_i; worst-case replacement of source i shifts θ by no more than R_i, and the union bound gives (6). The contaminated and clean marginal cumulative distribution functions differ in total variation by $1 - \prod_i(1 - \varepsilon_i)$; inverting under the lower density bound m yields the quantile statement. □

Practice. Set $\varepsilon_i \approx 0.05$ for widely-trusted demographic sources (Palestinian Central Bureau of Statistics, OHCHR) and $\varepsilon_i \approx 1$ (i.e., adversarial) for belligerent claims; (6) then absorbs the assumed political bias automatically.

6 Application: Gaza 2023–2026

We instantiate the framework using only inputs whose measurement is not controlled by either belligerent: PCBS population structure (Palestinian Central Bureau of Statistics, 2023–2024);

OCHA and Gaza MoH total deaths (United Nations Office for the Coordination of Humanitarian Affairs, 2025–2026); OHCHR and MoH demographic death records (United Nations Office of the High Commissioner for Human Rights, 2024); Lancet excess-mortality information (Khatib et al., 2024, 2025; Spagat et al., 2026); and IISS / RUSI / CSIS combatant-strength estimates (International Institute for Strategic Studies, 2024; Royal United Services Institute, 2024). The IDF claim itself (Israel Defense Forces, 2024–2026) is the object of the test, not an input.

Table 2. Diagnostic inputs for Gaza. The disputed quantity D_M is tested against the other rows.

Input	Value	Channel
w	0.733	PCBS population structure
a	0.255	PCBS age pyramid
ω	0.693 (OHCHR), 0.560 (MoH), 0.620 blend	identified deaths
f	0.0157–0.0269	Hamas/PIJ pre-war manpower / population
D	$\approx$ 70,000 direct deaths	OCHA/MoH, late 2025
$\bar{\mu}$	2.5	exposure-cap sensitivity parameter
claim	17–25k combatants	IDF public statements

Step 1: rejecting uniform random violence. The naive null "violence is uniform across the population" predicts $\mathbb{E}\omega = w = 73.3\%$. The OHCHR sample ($n = 8{,}119$, $\hat{\omega} = 69.3\%$) gives $z = -8.16$, $p \approx 1.7 \times 10^{-16}$, decisively rejecting uniformity. Adult males are overrepresented—necessarily, since the universe contains both combatants and structurally-exposed civilian men. Rejecting uniformity is necessary but not sufficient to rationalise any *specific* combatant share.

Step 2: identified set. On the blended anchor $\hat{\omega} = 0.620$,

$$\hat{q}_{\mu=1} = 1 - \tfrac{0.620\cdot(1-0.02)}{0.733} \approx 0.17, \qquad q \in [0,\, 0.17] \text{ at } \bar{\mu} = 2.5.$$

On the OHCHR anchor the upper endpoint is about 7% even at $\mu = 1$.

Step 3: claim test. The IDF range of 17–25k combatants over a 70k denominator gives $q^{\mathrm{IDF}} \approx 24$–$36\%$, far above $q(1)$. At $\bar{\mu} = 2.5$, rationalising it requires $\omega \in [0.39, 0.49]$, i.e., 13–23 points below the blended anchor. With $\sigma_\omega \approx 0.01$, this is roughly 20–23$\sigma$, dominating both other axes:

$$\rho_{\mathrm{IDF}} \gtrsim 20.$$

Step 4: spatial Bayesian posterior. Discretising Gaza into governorate cells, placing militants under a clustering prior, modelling strike intensity as a function of militant density, and estimating civilian adult-male exposure jointly with total deaths and the demographic records yields

$$q = 2.5\%, \qquad 95\% \text{ CrI} = [1.6\%, 3.8\%], \qquad \widehat{D_M} = 1{,}779 \; [1{,}112,\, 2{,}720].$$

The implied civilian-to-combatant ratio is 43:1 with a 95% credible interval $[30, 64]$. Figure 1 shows the full picture: panel A places the posterior anchors in the low-q region of the (ω, w) heatmap; panel B compares the posterior on q to the IDF claim band; panel C compares implied combatant counts on a log scale; panel D decomposes the ∼70k reported dead; panel E shows the civilian-to-combatant ratio. The point estimates align with the demographic Bayesian estimate of Gómez-Ugarte et al. (2025) once their reporting prior is set to ignore underreporting, and with the field-survey violent-death estimates of Spagat et al. (2026).

Symmetry. The test rejects the polar narrative $q = 0$ for the same reason: that value is also outside the feasible set $\mathcal{F}$ given the observed overrepresentation of adult males.

7 Empirical overview: 81 conflicts, 1900–2026

The contradiction-radius diagnostic requires demographic micro-data and manpower estimates, and cannot be applied sharply to every historical conflict. The 81-conflict dataset is used here only to map where civilian-share uncertainty is large, where indirect mortality dominates, and which conflicts would benefit most from identified-deaths samples; the supplement contains the per-conflict tables.

Generalisation. The diagnostic transfers to any conflict with (i) a population demographic decomposition, (ii) an independent demographic survey of the dead, (iii) an independent manpower estimate at t_0, and (iv) a reported aggregate death total. Such inputs exist for Tigray 2020–2022 (Ethiopian Statistical Service plus WHO surveys), Syria 2011– (Violations Documentation Center plus Syrian Network for Human Rights), the Mexican drug war (INEGI plus CSIS/IISS cartel estimates), the Iraq War 2003–2011 (Burnham et al., 2006), and Yemen 2014– (Yemen Data Project plus ACLED, Raleigh et al., 2010). Where these data are missing, the supplement marks the gap rather than inventing a contradiction radius.

8 Discussion

The conceptual contribution is the contradiction triangle (Theorem 4.3): no single number can be manipulated without forcing matching deviations in independently-measured quantities. The contradiction radius ρ then provides a unit-free measure of how far a claim sits from the consensus-feasible polytope. Whereas Spagat et al. (2009) frame war-death estimation as an *arena of contestation*, our framework gives the analyst a numerical referee for that arena, in the spirit of Lum et al. (2013) for multiple systems estimation and Radford et al. (2023) for Bayesian source aggregation.

Limitations.

1. Class AM is treated as a single bucket; finer age partitions (18–29, 30–49, 50+) tighten the bounds when age-resolved data exists.
2. Assumption 3.3 caps $\bar{\mu}$ a priori; if reality has $\bar{\mu} \to \infty$, only the manpower bound survives. Empirically $\bar{\mu} \leq 3$ in studied urban sieges.
3. The radius is conservative for non-Gaussian errors; a bootstrap or non-parametric posterior is preferred for small demographic samples.
4. The framework does not identify indirect deaths from blockade, famine, or healthcare collapse; these require separate excess-mortality methodology (Checchi and Roberts, 2008; Burnham et al., 2006; Degomme and Guha-Sapir, 2010; Gómez-Ugarte et al., 2025) and enter only via $\hat{D}$.

Reproducibility. Code, simulator, and inputs for the Gaza application are available at the companion repository.

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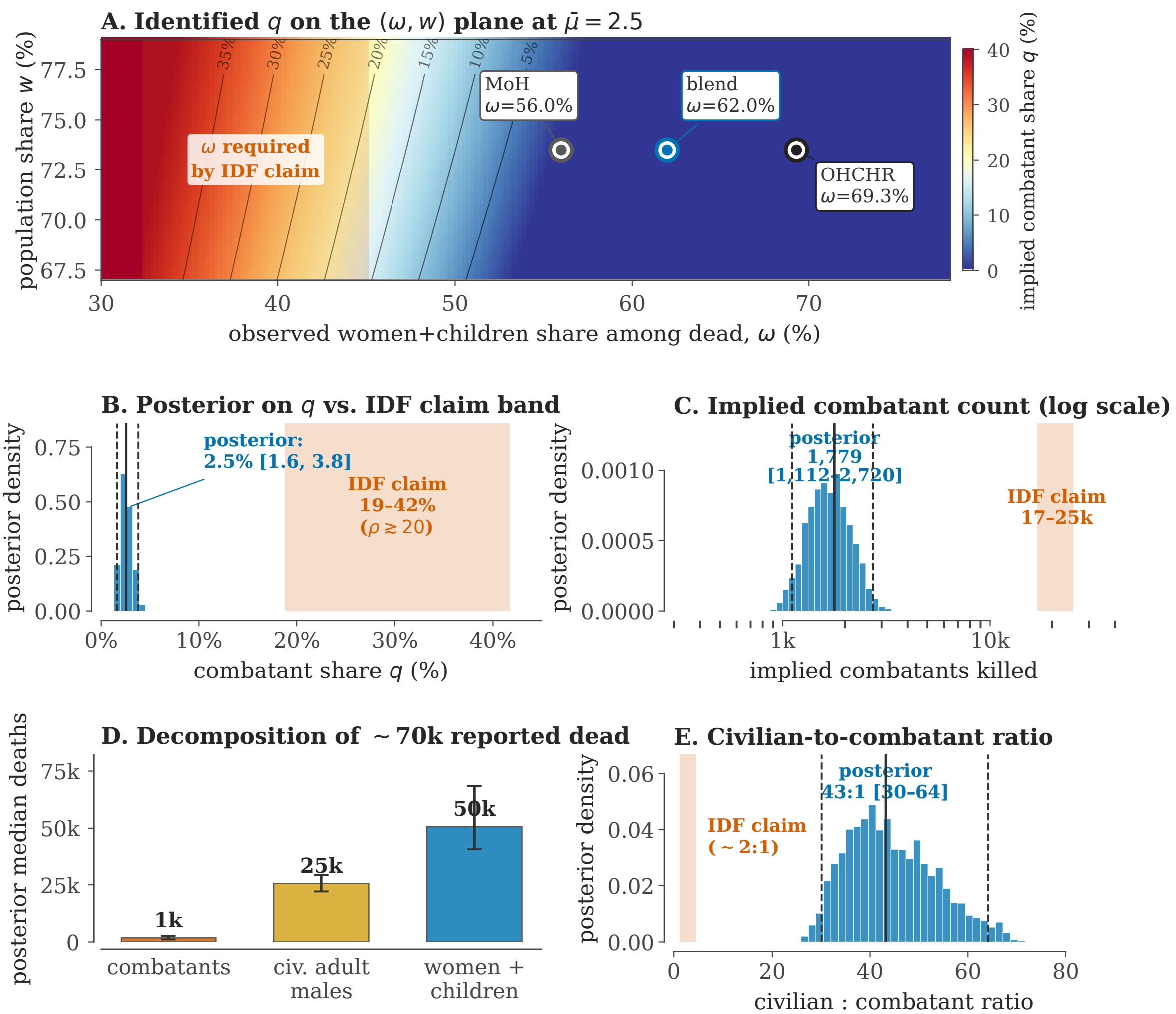

Figure 1. The diagnostic for Gaza (2023–2026). (A) Identified q on the (ω, w) plane at $\bar{\mu} = 2.5$. Iso-q contours are black; the OHCHR, blended, and Gaza Ministry of Health (MoH) anchors all sit in the low-q (blue) region; the IDF claim requires ω in the red band on the far left, far from any observed anchor. **(B)** Histogram of the spatial posterior on q (median, 95% credible interval as solid+dashed lines) vs. the IDF claim band converted to q. **(C)** Implied combatants killed (log axis) vs. the IDF claim of 17–25k. **(D)** Posterior decomposition of the ∼70k reported dead: ∼1k combatants, ∼25k civilian adult men, ∼50k women and children. **(E)** Civilian-to-combatant ratio: posterior median 43:1 vs. the IDF-implied ∼ 2:1.

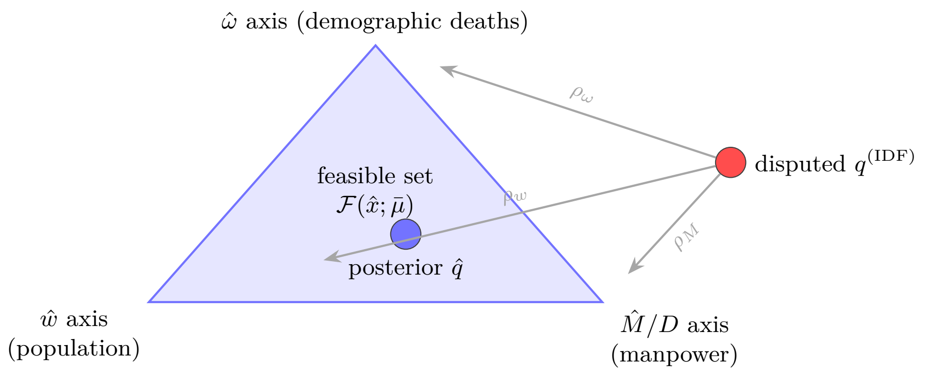

Figure 2. Contradiction triangle. The feasible set $\mathcal{F}$ is bounded by three independently measured "axes"; the contradiction radius $\rho = \min(\rho_\omega, \rho_w, \rho_M)$ is the shortest standardised perturbation from a disputed claim into $\mathcal{F}$.

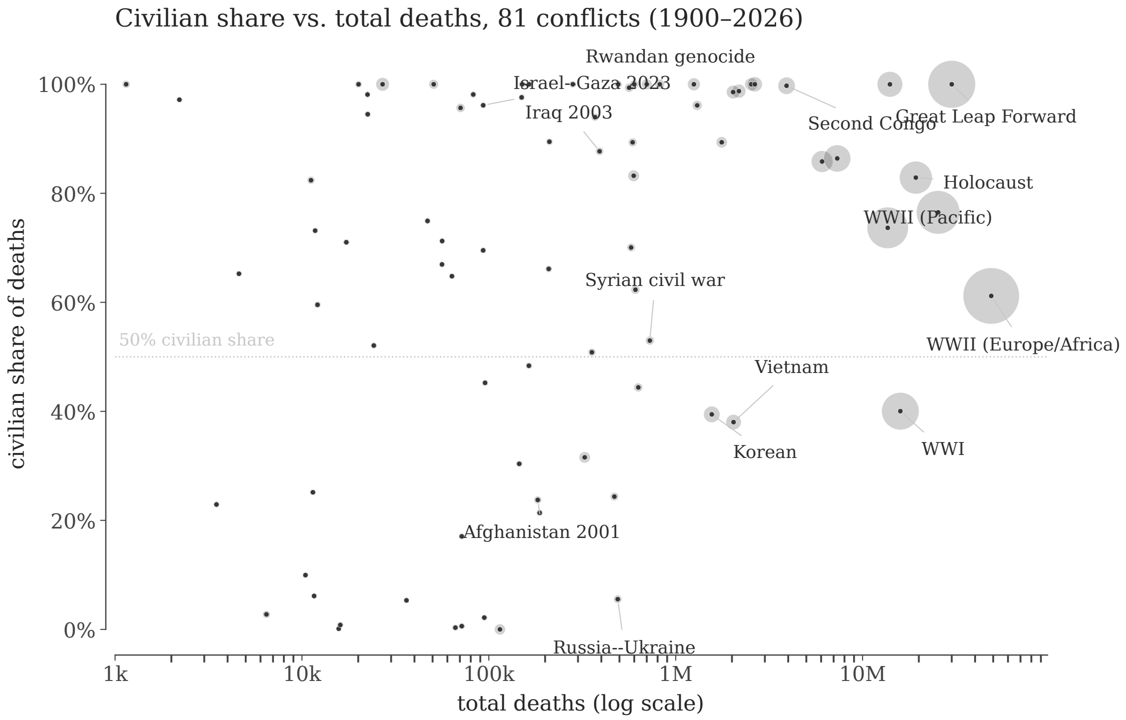

Figure 3. Civilian share vs scale, 1900–2026. Markers are conflicts; area is proportional to midpoint total deaths. Wars cluster bimodally: combat-dominated interstate wars (civilian share $\lesssim 30\%$) and civilian-targeting events, famines, and atrocities (civilian share $\gtrsim 90\%$).

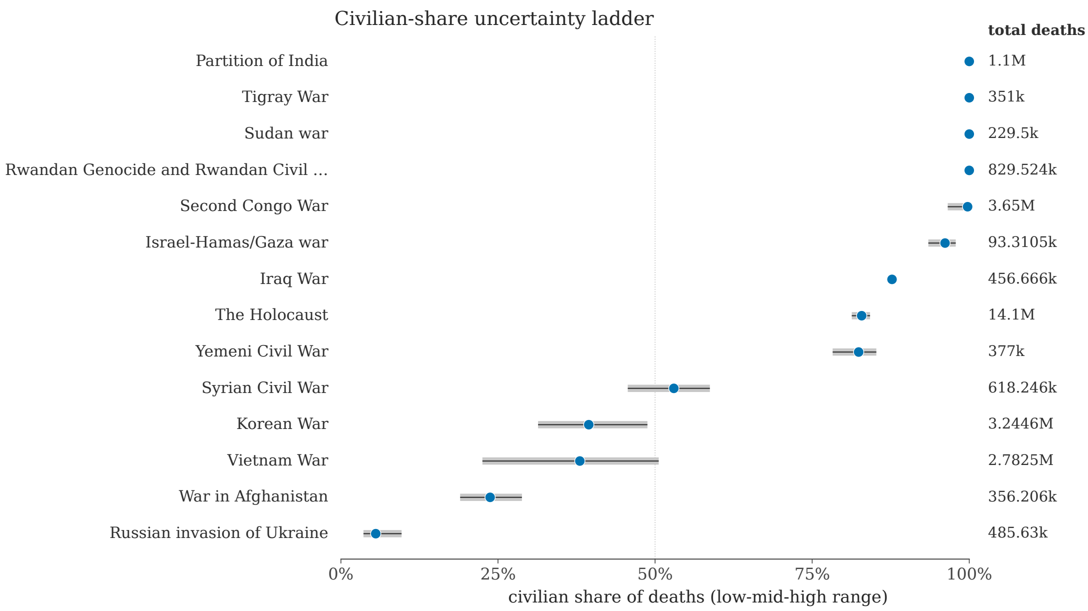

Figure 4. Civilian-share uncertainty ladder. Selected high-leverage conflicts. Grey segments are the side-level low/high implied civilian-share interval; blue dots are the midpoints. Wide intervals mark conflicts where the next technical advance is a better identified-deaths sample, not a better plot.